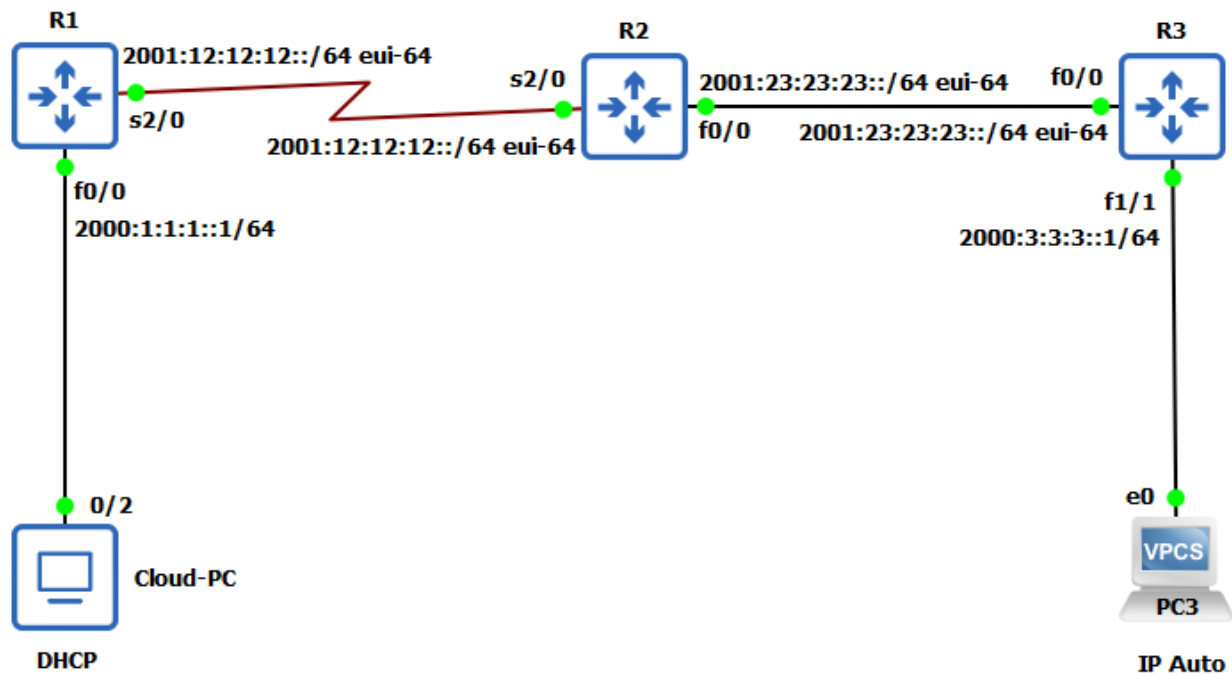


Default Routing – IPv6



Task

1. Configure routers with IPv6 address as shown in topology.
2. Configure IPv6 static routes (next-hop) on R2 router (ISP)
3. Configure default IPv6 routes (next-hop) on R1 & R3.
4. Cloud-PC & PC3 should be able to ping each other.

Note: Cloud-PC is a cloud which bridges the R1 router with loopback adapter on real PC.

R1 Router Configuration:

```

Router>enable
Router#config t
Router(config)#hostname R1
R1(config)#no ip domain-lookup
R1(config)#enable password ccna
R1(config)#ipv6 unicast-routing
R1(config)#int fa0/0
R1(config-if)#ipv6 address 2000:1:1:1::1/64
R1(config-if)#description Link to local LAN
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int s2/0
R1(config-if)#ipv6 address 2001:12:12:12::/64 eui-64
R1(config-if)#description Link to R2 on s2/0
R1(config-if)#encapsulation hdlc
  
```

```

R1(config-if)#clockrate 64000
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#line vty 0 15
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 1 0
R1(config-line)#exit
R1(config)#line console 0
R1(config-line)#password cisco1
R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 0 0
R1(config-line)#exit
R1(config)#exit
R1#

```

R2 Router Configuration:

```

Router>enable
Router#config t
Router(config)#hostname R2
R2(config)#no ip domain-lookup
R2(config)#enable password ccna
R2(config)#ipv6 unicast-routing
R2(config)#int s2/0
R2(config-if)#ipv6 address 2001:12:12:12::/64 eui-64
R2(config-if)#description Link to R1 on s2/0
R2(config-if)#encapsulation hdlc
R2(config-if)#clockrate 64000
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int fa0/0
R2(config-if)#ipv6 address 2001:23:23:23::/64 eui-64
R2(config-if)#description Link to R3 on fa0/0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#line vty 0 15
R2(config-line)#password cisco
R2(config-line)#login
R2(config-line)#logging synchronous
R2(config-line)#exec-timeout 1 0
R2(config-line)#exit
R2(config)#line console 0
R2(config-line)#password cisco1
R2(config-line)#login
R2(config-line)#logging synchronous
R2(config-line)#exec-timeout 0 0
R2(config-line)#exit
R2(config)#exit
R2#

```

R3 Router Configuration:

```

Router>enable
Router#config t
Router(config)#hostname R3
R3(config)#no ip domain-lookup
R3(config)#enable password ccna
R3(config)#ipv6 unicast-routing
R3(config)#int fa1/1
R3(config-if)#ipv6 address 2000:3:3:3::1/64
R3(config-if)#description Link to local LAN
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#int fa0/0
R3(config-if)#ipv6 address 2001:23:23:23::/64 eui-64
R3(config-if)#description Link to R2 on fa0/0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#line vty 0 15
R3(config-line)#password cisco
R3(config-line)#login
R3(config-line)#logging synchronous
R3(config-line)#exec-timeout 1 0
R3(config-line)#exit
R3(config)#line console 0
R3(config-line)#password cisco1
R3(config-line)#login
R3(config-line)#logging synchronous
R3(config-line)#exec-timeout 0 0
R3(config-line)#exit
R3(config)#exit
R3#

```

PC3 Configuration:

```

PC3> ip auto
GLOBAL SCOPE      : 2000:3:3:3:2050:79ff:fe66:6800/64
ROUTER LINK-LAYER : ca:03:1e:00:00:1d

```

Cloud-PC IP address:

Ethernet adapter Loopback:

```

Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2000:1:1:1:d5d6:56d2:75d0:b647
Temporary IPv6 Address. . . . . : 2000:1:1:1:a48d:f746:74b4:991f
Link-local IPv6 Address . . . . : fe80::d5d6:56d2:75d0:b647%24
IPv4 Address. . . . . : 192.168.1.10
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::c801:20ff:fe88:8%24
                          192.168.1.1

```

✓ Verification & Testing:

R1#sh int fa0/0 | include Hardware

Hardware is i82543 (Livengood), address is ca01.2088.0008 (bia ca01.2088.0008)
R1#

R1#sh ipv6 interface s2/0

Serial2/0 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::C801:20FF:FE88:8
No Virtual link-local address(es):
Description: Link to R2 on s2/0
Global unicast address(es):
2001:12:12:12:C801:20FF:FE88:8, subnet is 2001:12:12:12::/64 [EUI]
Joined group address(es):
FF02::1
FF02::2
FF02::1:FF88:8
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachable are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds (using 30000)
ND RAs are suppressed (periodic)
Hosts use stateless autoconfig for addresses.

R1#sh ipv6 interface fa0/0

FastEthernet0/0 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::C801:20FF:FE88:8
No Virtual link-local address(es):
Description: Link to local LAN
Global unicast address(es):
2000:1:1:1:1, subnet is 2000:1:1:1::/64
Joined group address(es):
FF02::1
FF02::2
FF02::1:FF00:1
FF02::1:FF88:8
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachable are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds (using 30000)
ND advertised reachable time is 0 (unspecified)
ND advertised retransmit interval is 0 (unspecified)
ND router advertisements are sent every 200 seconds
ND router advertisements live for 1800 seconds
ND advertised default router preference is Medium
Hosts use stateless autoconfig for addresses.

R1#

R2#sh int fa0/0 | include Hardware

Hardware is i82543 (Livengood), address is ca02.0c24.0008 (bia ca02.0c24.0008)
R2#

R2#sh ipv6 int s2/0

Serial2/0 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::C802:CFF:FE24:8
No Virtual link-local address(es):
Description: Link to R1 on s2/0
Global unicast address(es):
2001:12:12:12:C802:CFF:FE24:8, subnet is 2001:12:12:12::/64 [EUI]
Joined group address(es):
FF02::1
FF02::2
FF02::1:FF24:8
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachable are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds (using 30000)
ND RAs are suppressed (periodic)
Hosts use stateless autoconfig for addresses.

R2#sh ipv6 int fa0/0

FastEthernet0/0 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::C802:CFF:FE24:8
No Virtual link-local address(es):
Description: Link to R3 on fa0/0
Global unicast address(es):
2001:23:23:23:C802:CFF:FE24:8, subnet is 2001:23:23:23::/64 [EUI]
Joined group address(es):
FF02::1
FF02::2
FF02::1:FF24:8
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachable are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds (using 30000)
ND advertised reachable time is 0 (unspecified)
ND advertised retransmit interval is 0 (unspecified)
ND router advertisements are sent every 200 seconds
ND router advertisements live for 1800 seconds
ND advertised default router preference is Medium
Hosts use stateless autoconfig for addresses.

R2#

R3#sh int fa0/0 | include Hardware

Hardware is i82543 (Livengood), address is ca03.1e00.0008 (bia ca03.1e00.0008)
R3#

R3#sh ipv6 int fa0/0

FastEthernet0/0 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::C803:1EFF:FE00:8
No Virtual link-local address(es):
Description: Link to R2 on fa0/0
Global unicast address(es):
2001:23:23:23:C803:1EFF:FE00:8, subnet is 2001:23:23:23::/64 [EUI]
Joined group address(es):
FF02::1
FF02::2
FF02::1:FF00:8
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachable are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds (using 30000)
ND advertised reachable time is 0 (unspecified)
ND advertised retransmit interval is 0 (unspecified)
ND router advertisements are sent every 200 seconds
ND router advertisements live for 1800 seconds
ND advertised default router preference is Medium
Hosts use stateless autoconfig for addresses.

R3#sh int fa1/1 | include Hardware

Hardware is i82543 (Livengood), address is ca03.1e00.001d (bia ca03.1e00.001d)
R3#

R3#sh ipv6 int fa1/1

FastEthernet1/1 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::C803:1EFF:FE00:1D
No Virtual link-local address(es):
Description: Link to local LAN
Global unicast address(es):
2000:3:3:3::1, subnet is 2000:3:3:3::/64
Joined group address(es):
FF02::1
FF02::2
FF02::1:FF00:1
FF02::1:FF00:1D
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachable are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds (using 30000)
ND advertised reachable time is 0 (unspecified)

ND advertised retransmit interval is 0 (unspecified)
ND router advertisements are sent every 200 seconds
ND router advertisements live for 1800 seconds
ND advertised default router preference is Medium

Hosts use stateless autoconfig for addresses.

R3#

R1#sh ipv6 int brief

FastEthernet0/0 [up/up]
FE80::C801:20FF:FE88:8
2000:1:1:1::1
FastEthernet0/1 [administratively down/down]
unassigned
FastEthernet1/0 [administratively down/down]
unassigned
FastEthernet1/1 [administratively down/down]
unassigned
Serial2/0 [up/up]
FE80::C801:20FF:FE88:8
2001:12:12:12:C801:20FF:FE88:8
Serial2/1 [administratively down/down]
unassigned
Serial2/2 [administratively down/down]
unassigned
Serial2/3 [administratively down/down]
unassigned

R2#sh ipv6 int brief | exclude down

FastEthernet0/0 [up/up]
FE80::C802:CFF:FE24:8
2001:23:23:23:C802:CFF:FE24:8
unassigned
unassigned
Serial2/0 [up/up]
FE80::C802:CFF:FE24:8
2001:12:12:12:C802:CFF:FE24:8
unassigned
unassigned

R3#sh ipv6 int brief | exclude down|unassigned

FastEthernet0/0 [up/up]
FE80::C803:1EFF:FE00:8
2001:23:23:23:C803:1EFF:FE00:8
FastEthernet1/1 [up/up]
FE80::C803:1EFF:FE00:1D
2000:3:3:3::1

Static default routing on R1 Router

```
R1#config t
R1(config)#ipv6 route ::0 2001:12:12:12:C802:CFF:FE24:8
R1(config)#exit
R1#
```

Static routing on R2 Router

```
R2#config t
R2(config)#ipv6 route 2000:1:1:1::/64 2001:12:12:12:C801:20FF:FE88:8
R2(config)#ipv6 route 2000:3:3:3::/64 2001:23:23:23:C803:1EFF:FE00:8
R2(config)#exit
R2#
```

Static default routing on R1 Router

```
R3#config t
R3(config)#ipv6 route ::0 2001:23:23:23:C802:CFF:FE24:8
R3(config)#exit
R3#
```

✓ Verification & Testing:**R1#sh ipv6 route**

IPv6 Routing Table - default - 6 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

S ::0 [1/0]

via 2001:12:12:12:C802:CFF:FE24:8

C 2000:1:1:1::/64 [0/0]

via FastEthernet0/0, directly connected

L 2000:1:1:1::1/128 [0/0]

via FastEthernet0/0, receive

C 2001:12:12:12::/64 [0/0]

via Serial2/0, directly connected

L 2001:12:12:12:C801:20FF:FE88:8/128 [0/0]

via Serial2/0, receive

L FF00::/8 [0/0]

via Null0, receive

R1#

R2#sh ipv6 route static

IPv6 Routing Table - default - 7 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea
IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO
ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

```
S 2000:1:1:1::/64 [1/0]
  via 2001:12:12:12:C801:20FF:FE88:8
S 2000:3:3:3::/64 [1/0]
  via 2001:23:23:23:C803:1EFF:FE00:8
```

R2#

R3#sh ipv6 route static

IPv6 Routing Table - default - 6 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

```
S ::0 [1/0]
  via 2001:23:23:23:C802:CFF:FE24:8
```

R3#

R1#sh ipv6 protocols

IPv6 Routing Protocol is "connected"

IPv6 Routing Protocol is "ND"

IPv6 Routing Protocol is "static"

R1#

R2#sh ipv6 protocols

IPv6 Routing Protocol is "connected"

IPv6 Routing Protocol is "ND"

IPv6 Routing Protocol is "static"

R2#

R3#sh ipv6 protocols

IPv6 Routing Protocol is "connected"

IPv6 Routing Protocol is "ND"

IPv6 Routing Protocol is "static"

R3#

Ping PC3 from R1

```
R1#ping ipv6 2000:3:3:3:2050:79ff:fe66:6800
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:3:3:3:2050:79FF:FE66:6800, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 64/65/68 ms

R1#

Ping Cloud-PC from R3**R3#ping ipv6 2000:1:1:1:d5d6:56d2:75d0:b647**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:1:1:1:D5D6:56D2:75D0:B647, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 64/66/72 ms

R3#

Ping PC3 from Cloud-PC**C:\Users\Ratan> ping 2000:3:3:3:2050:79ff:fe66:6800**

Pinging 2000:3:3:3:2050:79ff:fe66:6800 with 32 bytes of data:

Reply from 2000:3:3:3:2050:79ff:fe66:6800: time=61ms

Reply from 2000:3:3:3:2050:79ff:fe66:6800: time=58ms

Reply from 2000:3:3:3:2050:79ff:fe66:6800: time=60ms

Reply from 2000:3:3:3:2050:79ff:fe66:6800: time=58ms

Ping statistics for 2000:3:3:3:2050:79ff:fe66:6800:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 58ms, Maximum = 61ms, Average = 59ms

Ping Cloud-PC from PC3**PC3> ping 2000:1:1:1:d5d6:56d2:75d0:b647**

2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=1 ttl=61 time=61.118 ms

2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=2 ttl=61 time=63.140 ms

2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=3 ttl=61 time=63.153 ms

2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=4 ttl=61 time=62.148 ms

2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=5 ttl=61 time=63.222 ms

Important Commands:

```
sh int xx | include Hardware
sh ipv6 interface xx
sh ipv6 int brief
sh ipv6 int brief | exclude down
sh ipv6 int brief | exclude down|unassigned
sh ipv6 route
sh ipv6 route static
sh ipv6 protocols
```

Ratan